

SYSTEMATIC REVIEW

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Strategies and prerequisites for combating health misinformation on social media: a systematic review

Leila Keikha¹, Azita Shahraki-Mohammadi^{2*} and Abdolahad Nabilahi³

Abstract

Objective The speed and complexity of transmitting health misinformation through social media can lead to the transfer of information that causes irreparable damage to the state of health, control, and prevention of diseases. This research aimed to identify the prerequisites and best strategies for combating health misinformation on social media.

Method The current systematic review was carried out following the PRISMA guidelines. In September 2024, a search was conducted using “misinformation” and “social media” keywords and their equivalents in selected databases (Scopus, Web of Science, and PubMed). Inclusion criteria comprised the implementation of an intervention aimed at combating health misinformation on social media, while studies not in English and those that did not address health misinformation on social media were excluded. The data were analyzed using the conventional content analysis method. EndNote 21 and Excel 2021 software were used to collect and analyze the articles.

Result Out of 6395 identified articles, 20 articles were included in the present study. Half of the studies addressing health misinformation were published in 2023 and 2024, with the United States leading the way. The combat of COVID-19 misinformation was the most frequent. From the content analysis of the included studies in a total of three strategies: communication strategies, technology-based strategies, and multimedia strategies to combat health misinformation on social media, it was identified. Four categories: needs assessment, educating community leaders, content design, and content quality assessment, were identified as the primary prerequisites to combat health misinformation on social media.

Conclusion Combating health information in social media requires basic infrastructures and the use of hybrid approaches. In addition, due to the different roles of celebrities and influencers, reputable health organizations and healthcare institutions should benefit from their participation in combating health misinformation.

Keywords Communication, Misinformation, Health, Strategy

*Correspondence:

Azita Shahraki-Mohammadi
azita.shahraki.m@gmail.com

¹Department of Library and Information Sciences, School of Allied Medical Sciences, Pregnancy Health Research Centre, Zahedan University of Medical Science, Zahedan, Iran

²Department of Medical Library and Information Sciences, School of Allied Medical Sciences, Health Promotion Research Center, Zahedan University of Medical Sciences, Boulevard Khalij-Fars, Zahedan, Sistan and Baluchestan 9816743175, Iran

³Department of Library and Information Sciences, School of Allied Medical Sciences, Research Institute of Cellular and Molecular Sciences in Infectious Diseases, Zahedan University of Medical Sciences, Zahedan, Iran



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Background

The trend of using social media is increasing day by day, and currently, about half of the world's population uses different types of this technology [1]. Surveys show that seven out of ten U.S. adults are posting about health-related issues, connecting to health-related groups, and following new health information [2]. False and misleading information and news may be distributed as genuine information and news in social media, and this misinformation, often created with various motives such as humor and political agendas, can be perceived as credible or real information by many people who lack sufficient knowledge and awareness [3]. The concepts of misinformation and disinformation differ in terms of their intention and method of dissemination. Misinformation includes information that is inadvertently incorrect and shared without the intention of harm, while disinformation includes false information that is deliberately created and shared to cause harm [4].

Advanced information and communication technologies, such as social media and the internet, have provided access to and facilitated the rapid dissemination of health information. On the other hand, the internet allows the dissemination of false and misleading health information that can lead to irreparable negative consequences [5]. On the other hand, social media can also facilitate support for health-related issues and empower patients, but the flood of misleading misinformation on social media is a major threat to public health [6]. In addition to confusion and reduced trust in health professionals, exposure to misinformation can delay effective care [4] and, in some cases [7], threaten people's lives [8]. Cases such as the anti-vaccine movement [7], the spread of natural cancer remedies [7], and rumors about infectious diseases, most of all, have occurred through social media during COVID-19.

As the use of social media was increasing, healthcare systems also faced various challenges. To address these challenges, some organizations have taken several measures to disseminate accurate and reliable information, such as fact-checking or false labeling. The increasing risk of health misinformation on social media has led to efforts to develop various interventions aimed at disseminating accurate information and maintaining public trust in evidence-based care. Identifying the type of actions in the field of combating health misinformation at different levels in social media can provide the ground for better management of health information content by health providers and policymakers. To the best of our knowledge, while previous studies have explored the issue of health misinformation, we do not find any studies that emphasize experimental intervention designs, particularly in searching for the implemented strategies and prerequisites. In this regard, the present study aimed to

investigate the prerequisites and strategies for combating health misinformation on social media.

Method

This systematic review study is reported by the PRISMA checklist. To identify relevant studies, the latest search was conducted on 18 September 2024 across three databases: PubMed, Web of Science, and Scopus. Moreover, no time constraints were taken into account when searching for sources in the selected databases. PubMed was chosen as the most authoritative source of health information to retrieve relevant documents, and to a large extent, the articles indexed in this database overlap with databases such as Embase. Additionally, Scopus and Web of Science were utilized to extract studies related to health misinformation because of their international reputation as citation databases.

The searched keywords and terms included Misinformation, Disinformation, and social media. This keyword, along with related keywords and synonyms that indicated false/fake information and social media, was searched. The search strategy in the different databases is provided in Table 1.

After removing duplicates, two researchers independently screened the titles and abstracts of the retrieved articles to identify relevant studies. Finally, for the included articles, the two researchers independently reviewed the full texts of the articles, and the findings were extracted based on the research purpose. Any disagreements were resolved through further discussion and the opinion of a third researcher. It should be noted that in this study, all types of false health information on social media, including fake news, rumors, and inaccurate health claims, are considered misinformation. End-Note version X9 Software was used to screen the articles. Additionally, Excel 2019 software was used to analyze the included articles. The data were analyzed using the conventional content analysis method provided by Graneheim and Lundman [9]. The current thematic analysis was performed using an inductive coding approach. In this manner, the research team utilized codes, categories, and themes derived from pertinent textual data on health misinformation, prevention strategies, and their essential requirements. The content analysis was conducted by two independent researchers (AN, LK), and the identified themes were categorized. To ensure the quality of themes, any disagreements were resolved with the input of a third researcher (ASM). To increase reliability, interrater agreement was evaluated by Kappa Cohen, which is a quantitative measure for evaluating consistency among reviewers. Extracted codes obtained a Kappa Cohen coefficient of 0.85, indicating a significant level of agreement [10].

Table 1 The search strategy in the three databases

Database	Search Strategy	Retrieved documents
PubMed	((("misinformation"[Title/Abstract] OR "disinformation"[Title/Abstract] OR "fake news"[Title/Abstract] OR "inaccurate information"[Title/Abstract] OR "poor quality information"[Title/Abstract] OR "low quality information"[Title/Abstract] OR "misleading information"[Title/Abstract] OR "rumor"[Title/Abstract] OR "gossip"[Title/Abstract] OR "hoax"[Title/Abstract] OR "urban legend"[Title/Abstract] OR "myth"[Title/Abstract] OR "fallacy"[Title/Abstract] OR "conspiracy theory"[Title/Abstract] OR "infodemic"[Title/Abstract]) AND ("Medical"[Title/Abstract] OR "health"[Title/Abstract])) OR ("disinformation"[MeSH Terms] AND "infodemic"[MeSH Terms])) AND ("Medical"[Title/Abstract] OR "health"[Title/Abstract]) AND ("Twitter"[Title/Abstract] OR "Facebook"[Title/Abstract] OR "Instagram"[Title/Abstract] OR "Flickr"[Title/Abstract] OR "YouTube"[Title/Abstract] OR "Reddit"[Title/Abstract] OR "Myspace"[Title/Abstract] OR "Pinterest"[Title/Abstract] OR "WhatsApp"[Title/Abstract] OR "social media"[Title/Abstract] OR "virtual space"[Title/Abstract] OR "Facebook"[Title/Abstract] OR "telegram"[Title/Abstract] OR "social media"[MeSH Terms]); Search date: 11/09/2024	1762
Web of Science	((AB=(misinformation OR disinformation OR "fake news" OR "inaccurate information" OR "poor quality information" OR "low-quality information" OR "misleading information" OR "rumor" OR "gossip" OR "hoax" OR "urban legend" OR "myth" OR "fallacy" OR "conspiracy theory" OR infodemic*)) AND AB=(medical OR health)) AND AB=(Twitter OR Facebook OR Instagram OR Flickr OR YouTube OR Reddit OR Myspace OR Pinterest OR WhatsApp OR "social media" OR "virtual space" OR Facebook OR Telegram) Search date: 11/09/2024	1611
Scopus	TITLE-ABS-KEY (misinformation OR disinformation OR "fake news" OR "inaccurate information" OR "poor quality information" OR "low-quality information" OR "misleading information" OR "rumor" OR "gossip" OR "hoax" OR "urban legend" OR "myth" OR "fallacy" OR "conspiracy theory" OR infodemic*) AND TITLE-ABS-KEY (medical OR health) AND TITLE-ABS-KEY (twitter OR Facebook OR Instagram OR Flickr OR YouTube OR Reddit OR Myspace OR Pinterest OR WhatsApp OR "social media" OR "virtual space" OR Facebook OR telegram) AND (LIMIT-TO (DOCTYPE, "ar")) Search date: 11/09/2024	3022

Although the research team attempted to prevent overlap between categories during the coding stage, it remains possible that overlap may occur in categories like communication and multimedia strategies. To reduce this overlap when analyzing the included articles to extract strategies to combat medical misinformation, the main method presented in the articles was considered the main method of combating this information.

Inclusion criteria

The key inclusion criterion for articles in the study was the implementation of an intervention aimed at combating health misinformation on social media. Additionally, studies that discussed experiences related to combat health misinformation were also included in the research.

Exclusion criteria

Studies that were not in English or whose full text was not accessible were excluded from the study. We also excluded retracted articles and conference papers. Original articles that did not present misinformation outside of health contexts or interventions aimed at addressing health information were excluded from the study. Additionally, the current study did not examine grey literature, reports, reviews, short articles, editorials, letters to the editor, and commentaries.

Results

A total of 6,395 articles were identified from the search of the three databases. Of these, 2,952 were duplicates and removed, leaving 3,443 articles. After screening the

titles and abstracts, 993 articles were eliminated. Of the remaining 2,504 documents, 2,484 records were excluded for reasons such as not addressing health misinformation, not being original, and Scientometrics, retracted articles, and others, resulting in 20 articles. The study selection process was illustrated in Fig. 1.

As shown in Table 2, the descriptive characteristics of the articles indicate that more than half of the 20 selected articles were published between 2023 and 2024. The United States was the most frequent country publishing studies aimed at combating health misinformation on social media. COVID-19 misinformation emerged as the most common topic, while Twitter, Facebook, and WhatsApp were the primary platforms used to address health misinformation. Additionally, most studies assessing the impact of combating misinformation employed a pre-/post-test design.

Strategies to combat health misinformation on social media

From the content analysis of 20 studies included in the present study, a total of three communication strategies, technology-based strategies, and multimedia strategies were identified to combat misinformation in social media. Details are provided in Table 3.

Communication strategies

Several studies have employed communication strategies, including those involving community leaders such as religious leaders, celebrities, influencers, and peers; community scientists; and community-based organizations

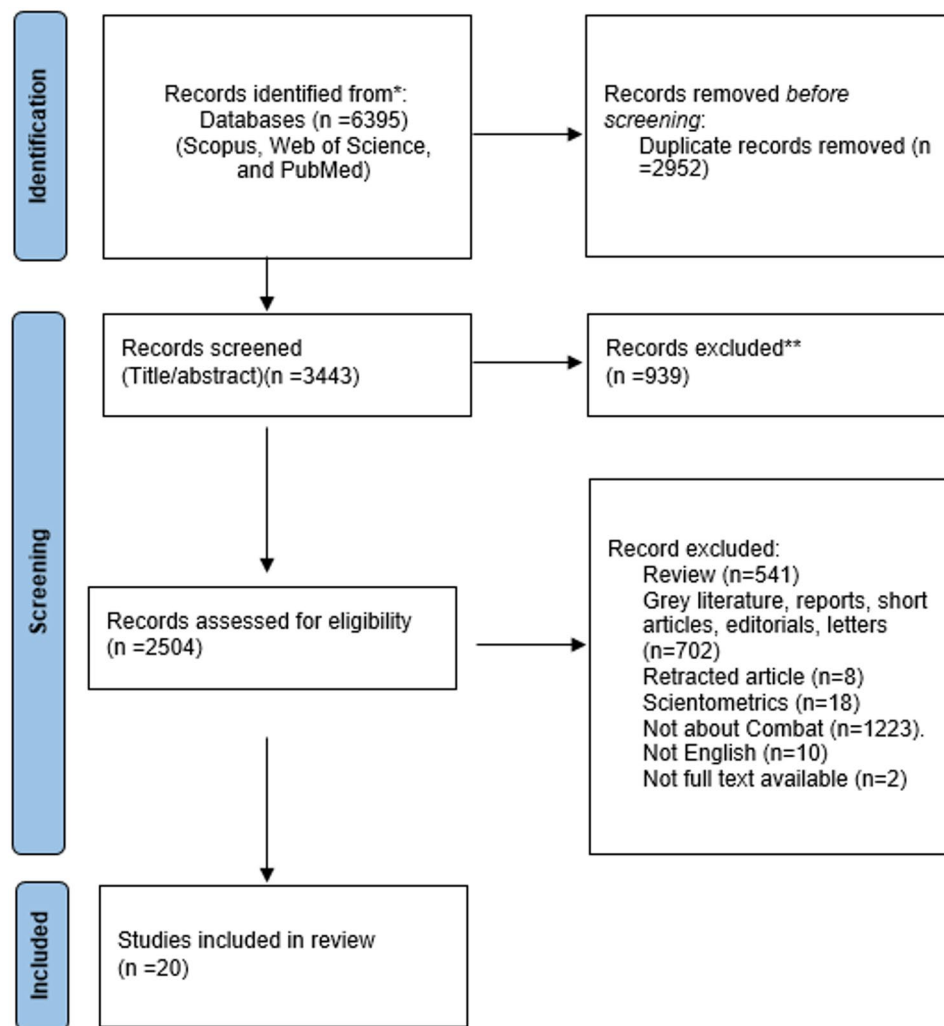


Fig. 1 Flow diagram of the study selection process

[14–16, 23, 26, 27, 29], to combat health misinformation on social media.

Technology-based strategies

Although technology was employed in the majority of methods of combating health misinformation on social media, several of the reviewed studies utilized strategies based on advanced technologies, including automated fact-checking/deleting methods, false labeling [22, 25], and prompt-based curriculum [17]. The relative effectiveness of journalistic fact checks that provide factual information in countering false claims, compared to the more commonly used simple fake news labels, showed that the lie labels had little impact on people's accuracy ratings and social media sharing [21].

Multi-media strategies

Most of the included studies employed hybrid methods and utilized multimedia to combat health

misinformation. Additionally, in most of these studies, evidence-based and reliable sources in various content formats, such as videos, text messages, pamphlets, and infographics, were used [12, 15, 21, 23, 24]. In one study, a serious game was employed to prevent and control health rumors in the context of the COVID-19 disease [11].

Prerequisites for combating health misinformation on social media

According to Table 4, in the 20 studies included in this thematic analysis, four categories- needs assessment, educating community leaders, content design, and content quality assessment- were identified as the primary prerequisites for implementing interventions to combat health misinformation on social media.

Table 2 Descriptive characteristics of the included articles ($n=20$)

No	Author	Year	Study design	Country	Sample	Subject	Platform
1	Xiong et al. [11]	2024	Experimental	China	200 elderly	COVID	WeChat
2	Alsaad E, AlDossary S [12]	2024	Experimental	Saudi Arabia	483	General health	WhatsApp
3	Ugarte, D. A., & Young, S [13]	2023	Experimental	USA	120 participants & 12 leader	COVID-19	Facebook
4	Silesky, Melissa Dunn ... et al [14]	2023	Mixed-method	USA	50 influencer posts	COVID-19	Instagram, Facebook, and Twitter
5	Kerrigan V et al. [15]	2023	Experimental	Australia	35 participants, First Nations adults	COVID-19	Facebook, Video, YouTube, Twitter, and LinkedIn are websites
6	Marx J et al. [16]	2023	Mixed-method	Brazil	536 tweets, 24 profiles of public health organizations	COVID-19	twitter
7	Peng Z et al. [17]	2023	experimental	China	320 participants	COVID-19	twitter
8	White et al. [18]	2023	Mixed-method	Different countries (30)	215,469,045 social posts	COVID-19	Twitter, Facebook, and other web content
9	Myneni et al. [19]	2023	Mixed-method	USA	Scientific community	COVID-19	Facebook, Twitter, and YouTube
10	Qasim R et al. [20]	2023	Mixed-method	Pakistan	20+35	COVID-19	WhatsApp, YouTube, and Facebook
11	Kim SJ et al. [21]	2022	Experimental	USA	995 Parents with children who are not vaccinated	HPV vaccine	MTurk or Facebook
12	SE Kreps, DL Kriner [22]	2022	Experimental	USA	2000 adult	COVID	Facebook
13	Jain et al. [23]	2022	experimental	USA	4000 health care 4000 followers	COVID-19	Facebook, Twitter, and Instagram
14	Barbaro A, et al.[24]	2021	Descriptive	Italy	None	General Health	Web portal(ISSalute) Twitter, Facebook
15	Kauk J...et al [25]	2021	Experimental	Germany	5611 Hashtag	COVID-19	twitter
16	Destaw Bayable. et al. [26]	2021	qualitative	Ethiopia	-	COVID-19	Mass media, Social media
17	Muñoz-Sastre, D ...et al [27]	2021	Mixed-method	Spain	849tweets	COVID-19	twitter
18	Xosé López-García ... et al. [28]	2021	Mixed-method	Spain	146 hoaxes	COVID-19	whatsapp
19	Bowles et al. [29]	2020	Experimental	Zimbabwe	-	COVID-19	whatsapp
20	M.ConnorSullivan [30]	2019	experimental	USA	625 participant	Flu vaccine	Facebook

Table 3 Strategies for combating health misinformation on social media

Items	Main Category	Subcategory
Combat Strategies	Communication strategies	Community Leaders
		Community Scientists
		Community-based organizations
	Technology-based strategies	False Labeling
		Fact-checking/Deleting
	Multi-media Strategies	Prompt-based curriculum learning Serious Game EBM (video/massage/pamphlet/infographic...etc)

Needs assessment

In several studies, interventions were designed to combat misinformation based on a needs assessment of the misinformation disseminated at the community level. Two studies targeted parents [21] and the broader public community [19]. In a study aimed at combating health misinformation, the World Health Organization (WHO) encouraged people to report misinformation on social

Table 4 Prerequisites for combating health misinformation on social media

Items	Main Category	Subcategory
Prerequisites for combating health misinformation	Needs assessment	Encourage people to report misconceptions
		Gather community needs
		Gather parents' needs.
		Social demand for fact-checking
	Educating community leaders	Famous person/Influencers
		Peers Leaders
		Religious Leaders
		Public Leaders
	Design content	Use expert designers
		Reliable sources
		Well-structured content
		Cultural norms
	Content quality assessments	Native languages
		Lingual experts
		Scientific experts

media [27]. A study also mentioned social demand as a criterion for conducting fact-checking interventions on health misinformation on social media [28].

Educating community leaders

In most studies that utilized community leaders to implement interventions combating health misinformation, the leaders received training to enhance their health literacy before the intervention. This training equipped them to educate community members, respond to questions, exchange ideas, and produce relevant content [13, 23, 26].

Design content

In several studies reviewed before the intervention, actions and considerations for content design were mentioned. These studies utilized experts, including videographers, graphic designers, photographers, and IT and communications specialists, to design and create content [15, 24]. Additionally, attention was paid to the use of reliable and evidence-based sources in the content. A study examined a structured template for delivering consistent messages to combat health misinformation [15]. Several studies specifically designed for multilingual communities have focused on the use of native languages and the cultural and social norms within the community [14–16, 26].

Content quality assessments

In several studies, content quality was evaluated after content design. In these studies, two types of evaluation were identified: scientific content evaluation by scientists [15] and linguistic evaluation by language experts [21].

Discussion

Today, the dissemination of health misinformation on social media can lead to serious risks; actually, Incorrect health information provides the basis for incorrect decisions about one's health, increased anxiety and stress, especially in health crises, physical harm, and reduced trust to health professionals and reliable sources [15, 16, 26, 31–33]. In this regard, the present study aimed to investigate the methods and basic prerequisites to combat health misinformation on social media. The review of evidence showed that several methods, including community-based strategies, technology-based procedures, and multimedia solutions, have been used in different periods to deal with health misinformation.

Communication strategies in combating health misinformation

Studies have shown that designing a local strategy to combat misinformation and misconceptions at the community level with full participation of the people and

their stakeholders is effective [34–36]. The interventions approach that used collective participation of the community members, such as influential people, religious leaders, and healthcare workers, not only has better acceptance among end users but also has long-term effects, especially for combating misinformation and rumors about vaccination [20]. Silesky expressed that a Multi-pronged campaign involving various organizational groups, using mass media, influencers, celebrities, and volunteers, has been successful in identifying and addressing health misinformation [14].

In the less developed areas where people have limited access to health information services, religious leaders help raise public awareness and support preventive methods against common diseases such as AIDS [37]. In Rotolo's study, the expert community strategy was employed, involving groups of trainees and physicians to provide training and counter misinformation, thereby fostering positive perceptions of vaccination [38]. Other researchers have also emphasized that utilizing the expert community strategy, such as the nursing community, which is on the front lines of patient care, presents a significant opportunity to support and promote important public health messages and dispel doubts related to COVID-19 vaccination [37]. Healthcare providers are uniquely positioned to use their expertise and public trust to combat misinformation [39]. Additionally, influencers who are popular as modern leaders on social media can disseminate credible health information and improve the health literacy of their audiences [40]. In a scoping review study, the results showed that although influential people on social media can positively influence health behaviors, they can also carry the risk of spreading false health information [41]. Therefore, using new methods such as AI to identify the authenticity of influential people on social media can be effective in combating health misinformation [40]. It is important to highlight that the community-based strategy emphasizes a hybrid model that integrates community involvement and the real-world environment to combat health misinformation. This hybrid approach employs public health principles, community engagement, and evidence-based methods to address the spread of health misinformation [16]. Its goal is to build trust, enhance access to reliable resources, and improve the effectiveness of interventions by collaborating with local groups. Several key elements identified in studies for implementation in the community setting using a hybrid approach include comprehensive community engagement, the use of truth-driven information, co-creation of messages, and multi-channel communication to leverage social media, local media, and in-person educational campaigns against health misinformation [41, 42]. In fact, an effective strategy for addressing health information in diverse communities

necessitates culturally tailored messaging, the use of simple language and personal narratives, and the opportunity to foster two-way interaction and synergy among local doctors, religious leaders, educators, and community influencers.

Technology-based strategies

The spread of health misinformation across the community can disrupt the process of proper treatment for individuals. Therefore, technology-based methods have been designed to fact-check health misinformation. For example, there are institutions around the world to verify false health information [43]. Fact-checking websites such as 'Weibo refutes rumors' and 'Tencent seeks truth' are also working to refute and combat false health information [44]. In a study, López García et al. examined the accuracy of journalistic content during the COVID-19 pandemic using the Newtral and Maldita platforms [28]. Verification through platforms can serve as an effective tool for correcting misinformation and a desirable solution for blocking health misinformation [29]. Additionally, the labeling false health information is a simple and convenient way to debunk false claims that are easily seen online by people who don't check the accuracy of the information [45]. It should be noted that, because health information verification methods are a time-consuming process that requires medical expertise, automated verification methods have recently received attention from researchers [46].

Other technology-based methods for verifying the accuracy of health information including text mining, natural language processing, machine learning techniques, and content similarity measurement algorithms help to content verification and detecting misinformation [47]. [46, 48] For example, the EARS tool used machine learning algorithms to group social media posts using artificial intelligence-based tools by the World Health Organization during the COVID-19 pandemic. The platform categorized incorrect narrative information in everyday conversations into five main groups (the cause of the virus, illness, treatment, interventions (including prevention), and information and misinformation), providing a framework for assessing and detecting misinformation [18]. Other researchers have also identified real-time corrections, crowd-sourced fact-checking, and algorithmic tagging as effective technological strategies for combating inaccurate health information [49]. Additionally, the use of artificial intelligence tools such as ChatGPT in information management among specific groups, such as the elderly, may increase individuals' ability to detect misinformation [50].

Multi-media strategies in combating health misinformation

A third effective strategy to combat health misinformation is the use of various multimedia tools. Kim et al. found that using messages based on evidence derived from scientific articles and reputable health organizations such as the National Cancer Institute, the Cancer Society, and the CDC was effective in combating incorrect myths and changing people's behavior towards HPV vaccination [21]. A hybrid approach to countering health misinformation may utilize various community, technology, or multimedia strategies to achieve effectiveness in a shorter time frame. In fact, within a hybrid strategy, one or more of the identified methods may be employed simultaneously or over a sequential period. As an example, the study by Barbaro et al. employed a hybrid approach utilizing various tools, including text, video, storytelling, and graphics, to combat health misinformation. It stated that more humanized content, in relation to the real community environment, had a more significant impact [24]. In addition, Kerrigan et al. used an approach to developing evidence-based video content and Q&A discussions among Australian audiences to address misinformation about COVID-19 [15]. In one study, which used video to combat health misinformation, it was used to make videos more effective in two forms: without words, using gestures, music, images, and short interviews with experts [39].

Moreover, Researchers believe that serious games are another effective way to combat fake health information and misinformation in cyberspace, especially in the vaccination aspect [49, 51]. Serious games are a type of game that is created for a specific purpose other than entertainment. Foxman states that serious games are designed as a means of managing public space for discussion about a specific topic, based on technology [50]. Serious games in an engaging environment provide individuals with the opportunity to receive quick and appropriate feedback regarding their judgments about rumors and misinformation about health [52]. In addition, they act as a convenient channel for health care providers to increase awareness and promote health literacy among the community [53]. In the experimental study conducted by Xiong, a serious game based on health rumors was designed, titled "Fight the Virus." This program attempted to differentiate between the health knowledge of older Chinese society and the rumors; it serves as a mechanism to prevent and control rumors related to COVID-19 [11]. Furthermore, other researchers have explored a hybrid approach utilizing multimedia to address misinformation, combining two or more opposing or complementary sources/methods that may incorporate visual data, user self-expression, and specialized information [54]. In a social context, implementing this strategy should also

involve gathering user feedback to enhance the accuracy and transparency of information sources and minimizing errors.

Essential prerequisites to combat misinformation on social media

One of the objectives of this study was to clarify the preliminary needs while dealing with health misinformation on social media. The results of the analysis showed that four key categories, including assessing the needs of the target community, designing information content, evaluating information quality, and training community leaders, were considered as preliminary prerequisites for designing interventions.

In the health system, evaluating the audience's needs has been taken into account to enhance health literacy before educational interventions. Understanding the needs of the audience has been considered a necessity to identify widespread false narratives of health information [55]. Some studies have used different methods of needs assessment to design interventions. For example, in the study by Barbro et al., a committee was created to monitor web content to identify and address misinformation about health [24]. In another study, virtual meetings were held with mothers to understand the information needs of mothers, and the most important barriers to refusing the HPV vaccine were identified, including false health information; consequently, appropriate interventions were designed based on that [21]. In designing interventions to combat health misinformation, using experts can lead to the production of content of high quality. In a study by Bobrow et al., experts such as graphic designers, photographers, videographers, IT and communications specialists were used to produce content for a website to combat health misinformation [24]. Various studies used accurate, scientific, and reliable sources, such as reputable medical institutions, international organizations, or peer-reviewed scientific sources, to design interventions to combat misinformation on social media [21, 29].

In the study by Alsaad and Aldossary, conducted interventions to improve the detection of health misinformation on WhatsApp, information for preparing educational videos was used from reliable sources such as recommendations from the official website to prevent the spread of health misinformation, the World Health Organization, and the CRAAP test, which is a tool for assessing the quality of social media resources [12]. Following Hugo's recommendations, communication principles, the social and cultural characteristics of the audience, such as literacy level and language preferences, should be taken into account in designing the educational content [56]. Bowles et al. recommended the production of podcasts in different languages to combat fake health information during the epidemic [40]. In another study to promote

the identification of health misinformation in Saudi Arabia, content was designed in the classical Arabic language that could be understood by a wide audience [12]. Additionally, in a study titled the fight against misinformation about the HPV vaccine by using a message, attention was also paid to the structural features of messages, such as message length and the structure [21]. In the third category, training leaders (Famous People/Influencers, Peer Leaders, Religious Leaders, and Public Leaders) was identified as a primary prerequisite for dealing with health information. Workshops and training of celebrities and religious leaders improved their knowledge on preventive measures against COVID-19 and also influenced their health behaviors. They were also effective in producing and sharing creative content and messages on preventive measures about COVID-19 and combating health misinformation [57]. Studies that were designed in the community level to combat health misinformation used leaders such as celebrities [16, 20], peers [46], or religious leaders [20, 58]. Leaders benefited from training in answering questions and creatively designing content and messages to share with their audiences [13, 16, 20, 58]. In the study by Wijesinghe et al., the results showed that training religious leaders to combat misinformation can be beneficial since they are in constant contact with health authorities and stay up-to-date on current health policies. In addition, this training facilitated the creation of networks (national, provincial, regional and community) to disseminate preventive methods in the field of COVID-19 [58]. In evaluating health information, studies indicated that health information should be understandable, in simple language, and scientifically validated to counter misinformation. Therefore, after content design, evaluating the quality of the produced messages is of great importance. In a study by Jung Kim et al. about HPV, the quality of the product was examined by experts in the areas of adolescent health, cancer prevention, and health communications [21]. In a study to combat fake COVID-19 health information, the quality of the content was assessed by media experts, health experts, and linguists [15]. In the study by Alsaad and Aldossary conducted in Saudi Arabia, seven experts fluent in Arabic evaluated the content of the messages in terms of spoken and written content [12]. Julian Marks found that understanding the content of health information about COVID-19, distributed in the form of multilingual tweets, requires the use of multilingual experts to develop appropriate verification tools [16]. Other similar studies have also recommended the use of validation tools and criteria such as discernment and compiling information content based on accuracy, credibility, timeliness, understandability, and relevance [59, 60].

Conclusion

The present study aimed to identify the prerequisites and desired strategies for combating health misinformation on social media. Most actions were taken between 2023 and 2024 in the United States. Twitter, Facebook, and WhatsApp were the most widely used social media platforms for combating misinformation. Although each study used a specific method and strategy to combat misinformation, the majority of studies reported the positive impact of measures to combat misinformation. The results of the studies in this period showed that the most misinformation and rumors were spread about vaccination, so health care providers should inform the public through various means to prevent the spread of misinformation and, as a result, prevent disease and promote health.

The studies applied interesting approaches that seem to be applicable and generalizable everywhere in the world. A community-based strategy can be implemented by local authorities or trusted individuals. In addition, when using a technology-based and multimedia strategy, it is better to provide the key infrastructure in the community. Furthermore, two approaches should be considered in the area of spreading misinformation and rumors. In the aspect of preventing the spread of health misinformation, various approaches can be used, such as using local and religious leaders, influencers, experts, and specialized organizations to provide scientific and credible information to increase health literacy. After spreading misinformation in the health field, healthcare providers should also use various approaches, for example, technology-based measures including artificial intelligence, machine learning, and serious games to identify misinformation and increase community health literacy.

In general, due to the widespread dissemination of inaccurate health information on various social media, the hybrid method requires combating misinformation. Moreover, due to the influential role of celebrities, religious leaders, reputable health organizations, and healthcare providers, their participation should be utilized in correcting health information and combating misinformation. Due to the significance of AI and limited research in this area, further studies are strongly recommended on the role of AI to combat misinformation.

Limitation

The first limitation of the current study was the exclusion of non-English studies and those for which full texts were unavailable. The second limitation was that only three databases were searched. Additionally, only original studies were included; shorter studies and reports were not reviewed. Another limitation is that we did not employ a specific tool to assess the quality of our articles. Furthermore, we did not prospectively register the review

protocol in a database such as PROSPERO. While registering a protocol can increase transparency and reduce potential bias, it is not a strict requirement for systematic reviews. We took careful steps to maintain scientific rigor by applying a well-defined search strategy, clear inclusion and exclusion criteria, and PRISMA-based reporting, which we believe support the reliability and validity of our results. Present limitations include the possibility that findings published in non-English language sources may differ significantly from the results of the current studies. Moreover, the selected databases may impact the comprehensiveness of the study and limit its generalizability.

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Authors' contributions

Conceptualizing the study: ASH; Writing/Original Draft: ASH, AN; Writing Review & Editing: ASH, LK; Methodology/Formal Analysis: LK, ASH; Project Administration: ASH. All authors reviewed the manuscript.

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Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

The Zahedan University of Medical Ethics ethically approved this study with the ethics code IR.ZAUMS.REC.1403.485.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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